

Plotting_individual_sectors.R

April 4, 2025

Contents

log_plot	1
not_log_plot	2
prep_plot_data	3
Index	5

log_plot	<i>Create a log10 visual</i>
----------	------------------------------

Description

This is a simple helper function for ease of use. It runs `prep_plot_data` and builds useful map visuals of the input data.

Usage

```
log_plot(  
  input,  
  title,  
  zlim_min = NULL,  
  zlim_max = NULL,  
  filename,  
  plot_directory,  
  domain,  
  County_Tigerlines,  
  State_Tigerlines,  
  Urban_Tigerlines  
)
```

Arguments

<code>input</code>	SpatRaster. Intended to be a SpatRaster of gridded methane emissions.
<code>title</code>	Optional character providing the main title for the plot. The plot will have no main title if not provided.
<code>zlim_min</code>	Optional numeric providing the minimum value for the colorscale of the plot, representing the log-scale gridded methane in nmol/m2/s. Will default to the minimum of the data if not provided.

zlim_max	Optional numeric equivalent to zlim_min, but for the maximum value.
filename	Optional character providing the output filename for the plot. Will default to a png with the name of the input if not provided, unless the plot is for an entire sector, rather than subsector. This will use the same default filename, but place it in a separate subfolder.

Details

This function is intended to be used when verbose = TRUE for CH4_inventory_build. It allows for reasonable default scales/titles or set ones, e.g., if wanting to hold the colorscale constant across multiple similar sectors/subsectors.

Value

This function returns nothing, but does produce a plot with the gridded methane data colored on a log scale with NA values set to black. All axes are clearly labeled and state, county, and, if relevant, focus city boundaries are overlaid in greys and white.

Author(s)

Joe Pitt, <madeup@wisc.edu>
 Kris Hajny, <blank@fake.edu>
 Israel Lopez-Coto, <test@test.edu>

Examples

```
log_plot(central_flux,filename="Wastewater_dom_central",
title="Domestic Wastewater -\n EPA total distributed using \nClean Watersheds Needs Survey")
```

not_log_plot	<i>Create a visual</i>
--------------	------------------------

Description

This is a simple helper function for ease of use. It builds useful map visuals of the input data.

Usage

```
not_log_plot(
  input,
  title,
  zlim_min = NULL,
  zlim_max = NULL,
  filename,
  plot_directory,
  domain,
  County_Tigerlines,
  State_Tigerlines,
  Urban_Tigerlines
)
```

Arguments

input	SpatRaster. Intended to be a SpatRaster of gridded methane emissions.
title	Optional character providing the main title for the plot. The plot will have no main title if not provided.
zlim_min	Optional numeric providing the minimum value for the colorscale of the plot, representing the gridded methane in nmol/m ² /s. Will default to the minimum of the data if not provided.
zlim_max	Optional numeric equivalent to zlim_min, but for the maximum value.
filename	Optional character providing the output filename for the plot. Will default to a png with the name of the input if not provided, unless the plot is for an entire sector, rather than subsector. This will use the same default filename, but place it in a separate subfolder.

Details

This function is intended to be used when verbose = TRUE for CH₄_inventory_build. It allows for reasonable default scales/titles or set ones, e.g., if wanting to hold the colorscale constant across multiple similar sectors/subsectors.

Value

This function returns nothing, but does produce a plot with the gridded methane data colored with NA values set to black. All axes are clearly labeled and state, county, and, if relevant, focus city boundaries are overlaid in greys and white.

Author(s)

Joe Pitt, <madeup@wisc.edu>
 Kris Hajny, <blank@fake.edu>
 Israel Lopez-Coto, <test@test.edu>

Examples

```
not_log_plot(septic_flux,filename="Wastewater_dom_septic_national",
             "Domestic Wastewater - Septic\n national EPA septic distributed using \ndeveloped open space/low inten
             global(min(septic_flux,septic_flux2),min),
             global(max(septic_flux,septic_flux2),max))
```

```
prep_plot_data
```

Prepare input data for a logarithmic plot

Description

This is a simple helper function for ease of use. It just takes the log₁₀ of the data and sets infinite values to NA.

Usage

```
prep_plot_data(input)
```

Arguments

input SpatRaster. Intended to be a SpatRaster of gridded methane emissions.

Details

This function is intended to be used when `verbose = TRUE` for `CH4_inventory_build`. It is called by `log_plot` to help build plots for each individual sector/subsector. Some of these, particular those for point sources, are more useful on a log scale.

Value

The log-scaled input data is returned, after removing infinite values.

Author(s)

Joe Pitt, <madeup@wisc.edu>

Kris Hajny, <blank@fake.edu>

Israel Lopez-Coto, <test@test.edu>

Examples

```
prep_plot_data(central_flux)
```

Index

`log_plot`, [1](#)

`not_log_plot`, [2](#)

`prep_plot_data`, [3](#)